

Topology and Manifolds

MH3600

Chapter 1: Overview and Preliminaries

Assoc. Prof. François J. M. Gay-Balmaz

Division of Mathematical Sciences
School of Physical and Mathematical Sciences
Nanyang Technological University

Outline

- 1 What is this course about?
- 2 Sets and functions
- 3 Relations
- 4 Finite and infinite sets. Cardinalities.
- 5 References

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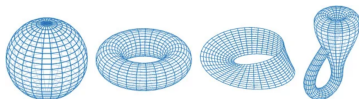
The main goal(s) of this course

We want to extend the ideas of mathematical analysis that we already learnt to **non-Euclidean spaces**. Topology and manifolds theory are the basis of **Global Analysis**.

Examples of non-Euclidean spaces:

• Spaces with non-trivial geometry:

- The sphere: $\mathbb{S}^2 := \{\mathbf{x} \in \mathbb{R}^3 \mid \|\mathbf{x}\| = 1\}$
- The torus: $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$
- Möbius strip, Klein bottle



• Matrix groups:

- The special linear group: $SL(n) := \{A \in \mathbb{M}(n) \mid \det(A) = 1\}$.
- The orthogonal group: $O(n) := \{A \in \mathbb{M}(n) \mid AA^T = \mathbb{I}\}$

How do we define continuity and differentiability on such spaces?

- **Function spaces:** they are typically infinite-dimensional
 - k -times continuously differentiable functions defined on an open subset $U \subset \mathbb{R}^n$ into $\mathbb{R}^m$: $C^k(U, \mathbb{R}^m)$.
 - Diffeomorphism group $\text{Diff}^k(\mathbb{R}^n)$.
- **Phase spaces in physics and robotics:**
 - Tangent bundles: used in Lagrangian mechanics

$$T\mathbb{S}^2 := \bigcup_{\mathbf{q} \in \mathbb{S}^2} T_{\mathbf{q}}\mathbb{S}^2 = \{(\mathbf{q}, \mathbf{v}) \in \mathbb{S}^2 \times \mathbb{R}^3 \mid \mathbf{v} \cdot \mathbf{q} = 0\}$$

is the phase space of the spherical pendulum.

- Cotangent bundles: used in Hamiltonian mechanics

$$T^*\mathbb{S}^2 := \bigcup_{\mathbf{q} \in \mathbb{S}^2} T_{\mathbf{q}}^*\mathbb{S}^2, \quad T_{\mathbf{q}}^*\mathbb{S}^2 = L(T_{\mathbf{q}}\mathbb{S}^2, \mathbb{R}).$$

How do we define continuity and differentiability on such spaces?

Given two sets M and N and a function $f : M \longrightarrow N$ between them:

- We need to define a **topological structure** on both on M and N in order to talk about the **continuity** of f .
- We need to define a **differentiable manifold structure** on both on M and N in order to talk about the **differentiability** of f .

Once we can talk about differentiability on manifolds, we can also formulate the analog of **differential equations (ordinary and partial)** using the notion of **vector field** and its **flow**.

All this is needed for the modeling of dynamical phenomena on spaces with non-trivial geometry.

Topological / differential invariants and classifications

A classical problem in topology/differential geometry is the identification of **invariants**. These are quantities that do not change under continuous/differentiable invertible transformations (that is **homeomorphisms/diffeomorphisms**).

An example of topological invariant is the **Euler characteristic** that for polyhedra is defined as

$$\chi = V - E + F,$$

where V , E , and F are the numbers of vertices (corners), edges and faces.

Any convex polyhedron's surface has Euler characteristic equal to 2 (Euler's Theorem).

Euler characteristic and orientability **completely classify closed surfaces** up to homeomorphism.

Theorem (Classification Theorem)

Any connected closed surface is homeomorphic to some member of one of these three families:

- *the sphere ($\chi = 2$),*
- *the connected sum of g tori for $g \geq 1$ ($\chi = 2 - 2g$),*
- *the connected sum of k real projective planes for $k \geq 1$ ($\chi = 2 - k$).*

Topology is also a very powerful language

Many important ideas in mathematical analysis, approximation theory, theory of differential equations, geometry, etc are naturally and efficiently formulated with the ideas introduced in this course.

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Sets and functions

These are the two main ingredients of mathematics in general and of mathematical analysis in particular.

We adopt the “naive” approach to set theory as in [Halmos] (i.e., defining sets and performing operations on them based on informal, intuitive notions).

Basic notation: sets are denoted using capital letters $A, B, \dots$ and their elements with lower case $a, b, \dots$.

- a belongs to A : $a \in A$.
- a does **not** belong to A : $a \notin A$.
- $A \subset B$: A subset of B when all the elements in A also belong to B .
- Subsets are usually determined by specification of a property. Braces are used in that case. Example: $B = \mathbb{N}$ and

$$A = \{x \in B \mid x \text{ is even}\} \subset B.$$

- Number sets: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{H}$.

Boolean operations

Union

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

Properties.

- Commutative: $A \cup B = B \cup A$.
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C$.
- Additional properties: $A \cup A = A$, $A \cup \emptyset = A$,

$$A \subset B \iff A \cup B = B.$$

Boolean operations (continued)

Intersection

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Properties.

- Commutative: $A \cap B = B \cap A$.
- Associative: $A \cap (B \cap C) = (A \cap B) \cap C$.
- Additional properties: $A \cap A = A$, $A \cap \emptyset = \emptyset$,

$$A \subset B \iff A \cap B = A.$$

- Distributive laws:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Boolean operations (continued)

Difference of two

Let A and B two sets. The **difference** $A \setminus B$ of A minus B (or the **complement** of B in A) is

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

DeMorgan's Laws.

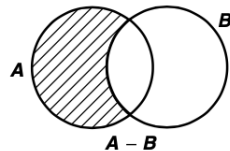
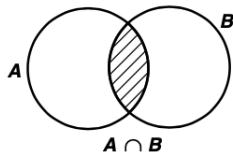
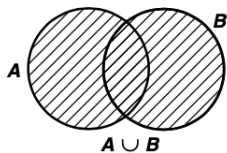
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

The **complement** of the **union** equals the **intersection** of the complements.

The **complement** of the **intersection** equals the **union** of the complements.

Taken from [Munkres]



Cartesian products

Another way of forming new sets from old ones using the notion of **ordered pairs**. The **Cartesian product** $A \times B$ is the set of all ordered pairs (a, b) for which a is an element of A and b is an element of B :

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Remarks:

- Not to confuse the ordered pair (a, b) with the open interval in the real line! Unfortunate duplicity in notation.
- This definition assumes that the concept of “ordered pair” is already given. It can be taken as a primitive concept, as was the notion of “set” or it can be given a definition in terms of the set operations already introduced. One definition in terms of set operations is expressed by the identity

$$\begin{aligned}(a, b) &= \{\{a\}, \{a, b\}\} \quad \text{when } a \neq b, \\(a, a) &= \{\{a\}\}.\end{aligned}$$

Functions

Definition

A **rule of assignment** is a subset r of the cartesian product $C \times D$ of two sets, having the property that each element of C appears as the first coordinate of at most one ordered pair belonging to r .

We think of r as a way of **assigning**, to the element c of C , the element d of D for which $(c, d) \in r$.

Given a rule of assignment r , the **domain** $\text{Dom}(r)$ of r is defined to be the subset of C consisting of all first coordinates of elements of r , and the **image** set $\text{Im}(r)$ of r is defined as the subset of D consisting of all second coordinates of elements of r . Formally,

$$\begin{aligned}\text{Dom}(r) &= \{c \in C \mid \text{there exists } d \in D \text{ such that } (c, d) \in r\} \\ \text{Im}(r) &= \{d \in D \mid \text{there exists } c \in C \text{ such that } (c, d) \in r\}\end{aligned}$$

Given a rule of assignment r , its domain and image are determined.

Definition

A **function** f is a rule of assignment r , together with a set $B \subset D$ that contains the image set of r . The domain $A \subset C$ of the rule r is also called the **domain** $\text{Dom}(f)$ of the function f . The image set of r is also called the **image set** $\text{Im}(f)$ of f , and the set B is called the **range** $\text{Range}(f)$ of f (also called **codomain** of f)

If f is a function having domain A and range B , we express this fact by writing

$$f : A \longrightarrow B$$

which is read “ f is a function from A to B ”, or “ f is a mapping from A into B ”.

Example

$$\begin{array}{ccc} f : \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & x^3 + 1 \end{array}$$

A function is not properly defined until **its three defining elements** have been specified: **domain, range, and rule of assignment**. The sentence “Let f be the function $f(x) = x^3 + 1$ ” is not a good function specification.

Restriction

Definition

If $f : A \longrightarrow B$ is a function and $A_0 \subset A$, the **restriction** $f|_{A_0} : A_0 \longrightarrow B$ is the function whose rule is the subset of $A_0 \times B$ given by

$$\{(a, f(a)) \mid a \in A_0\}$$

$f|_{A_0} : A_0 \longrightarrow B$ has been obtained out of $f : A \longrightarrow B$ by restriction of the **domain**. An analogous definition can (sometimes) be formulated using the restriction of the **range**.

EXAMPLE 1. Let $\mathbb{R}$ denote the real numbers and let $\bar{\mathbb{R}}_+$ denote the nonnegative reals. Consider the functions

$$\begin{array}{lll} f : \mathbb{R} \rightarrow \mathbb{R} & \text{defined by} & f(x) = x^2, \\ g : \bar{\mathbb{R}}_+ \rightarrow \mathbb{R} & \text{defined by} & g(x) = x^2, \\ h : \mathbb{R} \rightarrow \bar{\mathbb{R}}_+ & \text{defined by} & h(x) = x^2, \\ k : \bar{\mathbb{R}}_+ \rightarrow \bar{\mathbb{R}}_+ & \text{defined by} & k(x) = x^2. \end{array}$$

The function g is different from the function f because their rules are different subsets of $\mathbb{R} \times \mathbb{R}$; it is the restriction of f to the set $\bar{\mathbb{R}}_+$. The function h is also different from f , even though their rules are the same set, because the range specified for h is different from the range specified for f . The function k is different from all of these. These functions are pictured in Figure 2.1.

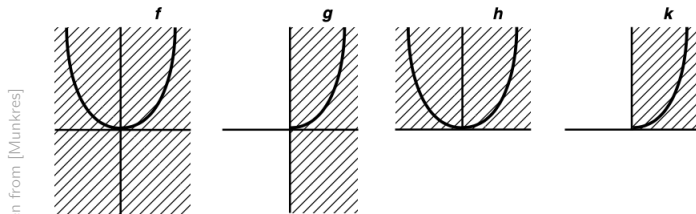


Figure 2.1

Taken from [Munkres]

Composition

Definition

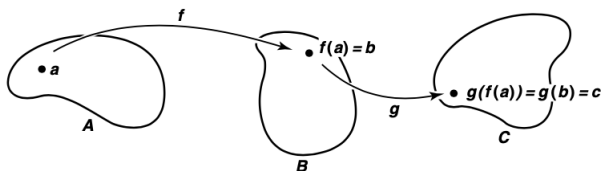
Given $f : A \longrightarrow B$ and $g : B \longrightarrow C$ define the **composition** of f and g as the function $g \circ f : A \longrightarrow C$ given by the identity

$$g \circ f(a) = g(f(a)).$$

$g \circ f : A \longrightarrow C$ is the function whose rule of assignment is

$$\{(a, c) \mid \text{for some } b \in B, f(a) = b \text{ and } g(b) = c\}.$$

Taken from [Munkres]



Injectivity and Surjectivity

Definition

Let $f : A \longrightarrow B$

- f is **injective or one-to-one** if for each pair of **distinct** points of A , their images under f are **distinct**:

$$f(a) = f(a') \implies a = a'.$$

- f is **surjective or onto** if every element in B is the image of an element of A under f :

$$b \in B \implies b = f(a) \text{ for at least one } a \in A.$$

- f is **bijective** when it is both **injective and surjective**.

Inverses, images, and preimages

Definition

If $f : A \longrightarrow B$ is **bijective** then there is another bijective map called the **inverse of f** $f^{-1} : B \longrightarrow A$ and defined by letting $f^{-1}(b)$ be the unique element a of A for which $f(a) = b$.

Using functions with restricted range, this definition can be extended to maps that are only **injective**.

Definition

Let $f : A \longrightarrow B$. If A_0 is a subset of A , we denote by $f(A_0)$ the set of all images of points of A_0 under the function f ; this set is called the **image of A_0 under f** . Formally,

$$f(A_0) = \{b \in B \mid b = f(a) \text{ for at least one } a \in A_0\}.$$

Definition

If B_0 is a subset of B , we denote by $f^{-1}(B_0)$ the set of all elements of A whose images under f lie in B_0 ; it is called the **preimage of B_0 under f** (or the **counterimage**, or the **inverse image**, of B_0). Formally,

$$f^{-1}(B_0) = \{a \in A \mid f(a) \in B_0\}.$$

The preimage $f^{-1}(B_0)$ is defined even if the function f is not injective.

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Relations

Definition

A **relation** on a set A is a subset C of the Cartesian product $A \times A$

If C is a relation on A , we use the notation xCy to mean the same thing as $(x, y) \in C$. We read it “ x is in the relation C to y ”.

Note that relations are more general than rules of assignment of functions $f : A \longrightarrow A$.

EXAMPLE 1. Let P denote the set of all people in the world, and define $D \subset P \times P$ by the equation

$$D = \{(x, y) \mid x \text{ is a descendant of } y\}.$$

Then D is a relation on the set P . The statements “ x is in the relation D to y ” and “ x is a descendant of y ” mean precisely the same thing, namely, that $(x, y) \in D$. Two other relations on P are the following:

$$B = \{(x, y) \mid x \text{ has an ancestor who is also an ancestor of } y\},$$

$$S = \{(x, y) \mid \text{the parents of } x \text{ are the parents of } y\}.$$

We can call B the “blood relation” (pun intended), and we can call S the “sibling relation.” These three relations have quite different properties. The blood relationship is symmetric, for instance (if x is a blood relative of y , then y is a blood relative of x), whereas the descendant relation is not. We shall consider these relations again shortly.

Taken from [Munkres]

Equivalence relations and partitions

Definition

A **equivalence relation** on a set A is a relation $C \subset A \times A$ having the following three properties:

- **Reflexive**: xCx for every x in A .
- **Symmetric**: If xCy , then yCx .
- **Transitive**: If xCy and yCz , then xCz .

Sometimes the tilde $\sim$ replaces C : $x \sim y$.

Definition

Given an equivalence relation $\sim$ on a set A and an element $x \in A$, we define a certain subset E_x of A , called the **equivalence class** determined by x , by the equation

$$E_x = \{y \in A \mid y \sim x\}.$$

- The equivalence class E_x determined by x contains x , since $x \sim x$.
- Two equivalence classes E_x and E_y are either disjoint or equal.
- The equivalence classes E_x , $x \in A$, associated to an equivalence relation C form a **partition** of A , that is, they are **disjoint** nonempty subsets whose **union** is all of A .
- Studying equivalence relations on a set A and studying partitions of A are really the same thing. Given any partition D of A , there is exactly one equivalence relation on A from which it is derived.

Examples

- (i) Define two points in the plane to be equivalent if they lie at the same distance from the origin. Reflexivity, symmetry, and transitivity hold trivially. The collection E_x of equivalence classes consists of all circles centered at the origin, along with the set consisting of the origin alone.
- (ii) Define two points of the plane to be equivalent if they have the same y -coordinate. The collection of equivalence classes is the collection of all straight lines in the plane parallel to the x -axis.
- (iii) Let $\mathcal{L}$ be the collection of all straight lines in the plane. Then $\mathcal{L}$ is not a partition of the plane, for distinct elements of $\mathcal{L}$ are not necessarily disjoint; two lines may intersect without being equal.

Order relations

Definition

A relation C on a set A is called an **order relation** (or a **simple order**, or a **linear order**) if it has the following properties:

- **Comparability**: For every x and y in A for which $x \neq y$, either xCy or yCx .
- **Nonreflexivity**: For no x in A does the relation xCx hold.
- **Transitivity**: If xCy and yCz , then xCz .

Relations that satisfy the last two conditions are called **strict partial order relations**.

Example: Consider the relation on the real line consisting of all pairs (x, y) of real numbers such that $x < y$. It is an order relation, called the **usual order relation**, on the real line. A less familiar order relation on the real line is the following: Define xCy if $x^2 < y^2$, or if $x^2 = y^2$ and $x < y$.

As the tilde $\sim$ is the generic symbol for an equivalence relation, the “less than” symbol $<$ is commonly used to denote an order relation. We shall use the notation $x \leq y$ to stand for the statement “either $x < y$ or $x = y$ ”. The notation $y > x$ stands for the statement $x < y$.

Definition

If X is a set and $<$ is an order relation on X and if $a < b$, we use the notation (a, b) to denote the set

$$\{x \mid a < x < b\}.$$

(a, b) is called an **open interval** in X . If this set is empty, we call a the **immediate predecessor** of b , and we call b the **immediate successor** of a .

- Suppose that A is a set ordered by the relation $<$. Let A_0 be a subset of A . We say that the element b is the **largest element** of A_0 if $b \in A_0$ and if $x \leq b$ for every $x \in A_0$. Similarly define the **smallest element** of A_0 . They are unique when they exist.
- We say that the subset A_0 of A is **bounded above** if there is an element b of A such that $x \leq b$ for every $x \in A_0$; the element b is called an **upper bound** for A_0 .
- If the set of all upper bounds for A_0 has a smallest element, that element is called the **least upper bound**, or the **supremum** of A_0 . It is denoted by $\sup A_0$; it may or may not belong to A_0 . If it does, it is the (unique) largest element of A_0 .
- Define similarly the **lower bounds** and the **greatest lower bound** or **infimum** of A_0 denoted by $\inf A_0$.

Definition

An ordered set A is said to have the **least upper bound property** if every nonempty subset A_0 of A that is bounded above has a least upper bound. Analogously, the set A is said to have the **greatest lower bound property** if every nonempty subset A_0 of A that is bounded below has a greatest lower bound.

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Finite and infinite sets. Cardinalities.

Definition

A set A is said to be **finite** if it is empty or if there is a bijection

$$f : A \longrightarrow \{1, \dots, n\}$$

for some positive integer n . In the former case, we say that A has **cardinality 0**; in the latter case, we say that A **has cardinality n** and we write $|A| = n$.

For instance, the set $\{1, \dots, n\}$ itself has cardinality n , for it is in bijective correspondence with itself under the identity function.

Important facts

The proofs of the following statements can be found in the textbook:

- (i) If A is finite then there is no bijection of A with a proper subset of itself.
- (ii) $\mathbb{Z}_+ = \{1, 2, 3, \dots\}$ is not finite: because the map $f : \mathbb{Z}_+ \longrightarrow \mathbb{Z}_+ \setminus \{1\}$ defined by $f(n) = n + 1$ is a bijection of $\mathbb{Z}_+$ with a proper subset of itself.
- (iii) The cardinality of a finite set A is uniquely determined by A .
- (iv) If B is a subset of the finite set A , then B is finite. If B is a proper subset of A , then the cardinality of B is less than the cardinality of A .
- (v) Finite unions and finite cartesian products of finite sets are finite.

Countable and uncountable sets

Definition

A set A is said to be **infinite** if it is not finite. It is said to be **countably infinite** if there is a bijective correspondence

$$f : A \longrightarrow \mathbb{Z}_+$$

A set is said to be **countable** if it is either finite or countably infinite. A set that is not countable is said to be **uncountable**.

- A subset of a countable set is countable.
- A countable union of countable sets is countable.
- A finite product of countable sets is countable.

Uncountable sets do exist

Given a set A denote by $\mathcal{P}(A)$ its **power set**

$$\mathcal{P}(A) = \{B \mid B \subseteq A\}.$$

Theorem

Let A be a set. There is no injective map $f : \mathcal{P}(A) \longrightarrow A$, and there is no surjective map $g : A \longrightarrow \mathcal{P}(A)$.

Proof

It suffices to prove the second statement because given a nonempty set B , the existence of an injective map $f : B \longrightarrow C$ implies the existence of a surjective map $g : C \longrightarrow B$.

Proof (continued)

To prove that a given map $g : A \longrightarrow \mathcal{P}(A)$ is **not surjective** note that for each $a \in A$ the image $g(a)$ of a is a subset of A which may or may not contain the point a itself. Let B be the subset of A consisting of all those points a such that $g(a)$ **does not contain** a :

$$B = \{a \mid a \in A \setminus g(a)\}.$$

We assert that B is a subset of A that does not lie in the image of g . For suppose that $B = g(a_0)$ for some $a_0 \in A$. We ask the question: Does a_0 belong to B or not? By definition of B ,

$$a_0 \in B \iff a_0 \in A \setminus g(a_0) \iff a_0 \in A \setminus B.$$

We have a contradiction. ■

Corollary

The set $\mathcal{P}(\mathbb{Z}_+)$ is not countably infinite.

Numerical equivalence

We define an equivalence relation $\sim$ in the space of non-empty sets called **numerical equivalence**:

$$A \sim B \iff \text{there exists a bijection between } A \text{ and } B.$$

- The numerical equivalence is an **equivalence relation**.
- The elements of each equivalence class associated to this relation are said to be **numerically equivalent** or that they have the same **cardinality/number of elements**.
- By the result cited above (see (iv) on slide 37) if $B \subset A$ and A is finite then B and A cannot have the same cardinality. This is **NOT true** when A is infinite.

Example: $\mathbb{Z}_+$, $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ have the same cardinality while $\mathbb{Z}_+ \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$. This common cardinality is denoted $\aleph_0$ (read aleph zero).

(proof of $\mathbb{Q} \sim \mathbb{Z}_+$ uses that if there is $g : \mathbb{Z}_+ \rightarrow B$ surjective then B countable, and uses $\mathbb{Z}_+ \times \mathbb{Z}_+$).

- By the Theorem on slide 39, for any set A :

$$|A| < |\mathcal{P}(A)|.$$

- By analogy with the finite cardinality situation, we denote $|\mathcal{P}(A)| = 2^{|A|}$.
- The cardinality of the real numbers is denoted by $\mathfrak{c}$ and called the **cardinality of continuum**. It can be proved that

$$\mathfrak{c} = 2^{\aleph_0}.$$

- We hence have the hierarchy

$$1 < 2 < 3 \cdots < \aleph_0 < \mathfrak{c} < 2^{\mathfrak{c}} < 2^{2^{\mathfrak{c}}} < \cdots$$

- $\aleph_0$ is the smallest infinite cardinal.
- The unsolved **Cantor's continuum hypothesis** states that there is no cardinal number greater than $\aleph_0$ and less than $\mathfrak{c}$.

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